

Title of Project: - Indian Defence Management System (SQL)

1. Aim of the Project:

The aim of this project is to design and implement a robust Database Management System (DBMS) to manage the operational data of the Navy and Army efficiently. Using Oracle as the backend database, this system will help in organizing, storing, retrieving, and managing information related to personnel, equipment, missions, and logistics. The project seeks to enhance data integrity, security, and quick access to critical information required for defence operations.

2. Description of the project in detail

This project involves creating a comprehensive database system tailored for the Navy and Army to streamline data management across various departments. The database includes tables for personnel records, ranks, equipment inventory, mission details, training schedules, and logistics management. Oracle's powerful features such as PL/SQL, triggers, and stored procedures are utilized to automate data validation, generate reports, and maintain referential integrity.

3. Applications of the project

- **Efficient Data Management:** Enables organized storage and retrieval of critical defense information.
- **Decision Support:** Provides commanders and staff with accurate and timely data to make informed decisions.
- **Resource Optimization:** Helps in tracking inventory and managing supplies effectively, reducing wastage.
- **Enhanced Security:** Ensures sensitive information is protected through Oracle's security features.
- **Operational Transparency:** Facilitates audit trails and reporting for accountability.
- **Training Management:** Keeps track of soldier training and certifications.

This system can also be adapted for use by other defense branches or security agencies for improved operational efficiency.

4. Code of the project

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```
CREATE TABLE Rank
(
rank_id      NUMBER PRIMARY KEY,
rank_name    VARCHAR2(50) NOT NULL
);

INSERT INTO Rank VALUES (1, 'Private');
INSERT INTO Rank VALUES (2, 'Sergeant');
INSERT INTO Rank VALUES (3, 'Captain');
INSERT INTO Rank VALUES (4, 'Major');
SELECT * FROM Rank;
```

Results Explain Describe Saved SQL History

0.00 seconds

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```
CREATE TABLE Rank
(
rank_id      NUMBER PRIMARY KEY,
rank_name    VARCHAR2(50) NOT NULL
);

INSERT INTO Rank VALUES (1, 'Private');
INSERT INTO Rank VALUES (2, 'Sergeant');
INSERT INTO Rank VALUES (3, 'Captain');
INSERT INTO Rank VALUES (4, 'Major');
SELECT * FROM Rank;
```

Results Explain Describe Saved SQL History

RANK_ID	RANK_NAME
1	Private
2	Sergeant
3	Captain
4	Major

4 rows returned in 0.00 seconds

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```

CREATE TABLE Unit
(
unit_id      NUMBER PRIMARY KEY,
unit_name    VARCHAR2(50) NOT NULL,
location     VARCHAR2(100) NOT NULL
);

INSERT INTO Unit VALUES (101, 'Northern Command', 'Jammu & Kashmir');
INSERT INTO Unit VALUES (102, 'Eastern Command', 'Assam');
INSERT INTO Unit VALUES (103, 'Naval Fleet Alpha', 'Mumbai');
SELECT * FROM Unit;

```

Results Explain Describe Saved SQL History

Table created.

0.00 seconds

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```

CREATE TABLE Unit
(
unit_id      NUMBER PRIMARY KEY,
unit_name    VARCHAR2(50) NOT NULL,
location     VARCHAR2(100) NOT NULL
);

INSERT INTO Unit VALUES (101, 'Northern Command', 'Jammu & Kashmir');
INSERT INTO Unit VALUES (102, 'Eastern Command', 'Assam');
INSERT INTO Unit VALUES (103, 'Naval Fleet Alpha', 'Mumbai');
SELECT * FROM Unit;

```

Results Explain Describe Saved SQL History

UNIT_ID	UNIT_NAME	LOCATION
101	Northern Command	Jammu & Kashmir
102	Eastern Command	Assam
103	Naval Fleet Alpha	Mumbai

3 rows returned in 0.00 seconds

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```
CREATE TABLE Soldier
(
soldier_id  NUMBER PRIMARY KEY,
name       VARCHAR2(50) NOT NULL,
age        NUMBER,
rank_id    NUMBER,
unit_id    NUMBER,
CONSTRAINT fk_soldier_rank FOREIGN KEY (rank_id) REFERENCES Rank(rank_id),
CONSTRAINT fk_soldier_unit FOREIGN KEY (unit_id) REFERENCES Unit(unit_id)
);

INSERT INTO Soldier VALUES (1001, 'Rahul Singh', 25, 1, 101);
INSERT INTO Soldier VALUES (1002, 'Arjun Mehta', 28, 2, 101);
INSERT INTO Soldier VALUES (1003, 'Vikram Rao', 32, 3, 102);
INSERT INTO Soldier VALUES (1004, 'Karan Kapoor', 40, 4, 103);
SELECT * FROM Soldier;
```

Results Explain Describe Saved SQL History

Table created.

0.02 seconds

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```
CREATE TABLE Soldier
(
soldier_id  NUMBER PRIMARY KEY,
name       VARCHAR2(50) NOT NULL,
age        NUMBER,
rank_id    NUMBER,
unit_id    NUMBER,
CONSTRAINT fk_soldier_rank FOREIGN KEY (rank_id) REFERENCES Rank(rank_id),
CONSTRAINT fk_soldier_unit FOREIGN KEY (unit_id) REFERENCES Unit(unit_id)
);

INSERT INTO Soldier VALUES (1001, 'Rahul Singh', 25, 1, 101);
INSERT INTO Soldier VALUES (1002, 'Arjun Mehta', 28, 2, 101);
INSERT INTO Soldier VALUES (1003, 'Vikram Rao', 32, 3, 102);
INSERT INTO Soldier VALUES (1004, 'Karan Kapoor', 40, 4, 103);
SELECT * FROM Soldier;
```

Results Explain Describe Saved SQL History

SOLDIER_ID	NAME	AGE	RANK_ID	UNIT_ID
1001	Rahul Singh	25	1	101
1002	Arjun Mehta	28	2	101
1003	Vikram Rao	32	3	102
1004	Karan Kapoor	40	4	103

4 rows returned in 0.00 seconds [CSV Export](#)

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```
CREATE TABLE Operation
(
operation_id  NUMBER PRIMARY KEY,
operation_name VARCHAR2(100) NOT NULL,
start_date   DATE,
end_date     DATE
);

INSERT INTO Operation VALUES (201, 'Operation Vijay', TO_DATE('1999-05-01', 'YYYY-MM-DD'), TO_DATE('1999-07-26', 'YYYY-MM-DD'));
INSERT INTO Operation VALUES (202, 'Operation Trident', TO_DATE('1971-12-04', 'YYYY-MM-DD'), TO_DATE('1971-12-06', 'YYYY-MM-DD'));
INSERT INTO Operation VALUES (203, 'Operation Parakram', TO_DATE('2001-12-15', 'YYYY-MM-DD'), TO_DATE('2002-10-16', 'YYYY-MM-DD'));
SELECT * FROM Operation;
```

Results Explain Describe Saved SQL History

OPERATION_ID	OPERATION_NAME	START_DATE	END_DATE
201	Operation Vijay	01-MAY-99	26-JUL-99
202	Operation Trident	04-DEC-71	06-DEC-71
203	Operation Parakram	15-DEC-01	16-OCT-02

3 rows returned in 0.00 seconds [CSV Export](#)

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```
CREATE TABLE Participation
(
  soldier_id    NUMBER,
  operation_id   NUMBER,
  role          VARCHAR2(50),
  CONSTRAINT pk_participation PRIMARY KEY (soldier_id, operation_id),
  CONSTRAINT fk_participation_soldier FOREIGN KEY (soldier_id) REFERENCES Soldier(soldier_id),
  CONSTRAINT fk_participation_operation FOREIGN KEY (operation_id) REFERENCES Operation(operation_id)
);

INSERT INTO Participation VALUES (1001, 201, 'Infantry');
INSERT INTO Participation VALUES (1002, 201, 'Medic');
INSERT INTO Participation VALUES (1003, 202, 'Commander');
INSERT INTO Participation VALUES (1004, 203, 'Naval Officer');
SELECT * FROM Participation;
```

Results Explain Describe Saved SQL History

SOLDIER_ID	OPERATION_ID	ROLE
1001	201	Infantry
1002	201	Medic
1003	202	Commander
1004	203	Naval Officer

4 rows returned in 0.00 seconds [CSV Export](#)

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```
CREATE TABLE Participation
(
  soldier_id    NUMBER,
  operation_id   NUMBER,
  role          VARCHAR2(50),
  CONSTRAINT pk_participation PRIMARY KEY (soldier_id, operation_id),
  CONSTRAINT fk_participation_soldier FOREIGN KEY (soldier_id) REFERENCES Soldier(soldier_id),
  CONSTRAINT fk_participation_operation FOREIGN KEY (operation_id) REFERENCES Operation(operation_id)
);

INSERT INTO Participation VALUES (1001, 201, 'Infantry');
INSERT INTO Participation VALUES (1002, 201, 'Medic');
INSERT INTO Participation VALUES (1003, 202, 'Commander');
INSERT INTO Participation VALUES (1004, 203, 'Naval Officer');
SELECT * FROM Participation;
```

Results Explain Describe Saved SQL History

Table created.

0.01 seconds

➤ Queries

- COMMIT

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```
INSERT INTO Participation VALUES (1001, 201, 'Infantry');
INSERT INTO Participation VALUES (1002, 201, 'Medic');
INSERT INTO Participation VALUES (1003, 202, 'Commander');
INSERT INTO Participation VALUES (1004, 203, 'Naval Officer');
SELECT * FROM Participation;

COMMIT;

SAVEPOINT before_update;

UPDATE Soldier SET age = 26 WHERE soldier_id = 1001;

ROLLBACK TO before_update;   -- undo update

DELETE FROM Soldier WHERE soldier_id = 1002;
ROLLBACK;   -- undo delete
```

Results Explain Describe Saved SQL History

Statement processed.

0.01 seconds

- **SAVEPOINT before update**

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```
INSERT INTO Participation VALUES (1001, 201, 'Infantry');
INSERT INTO Participation VALUES (1002, 201, 'Medic');
INSERT INTO Participation VALUES (1003, 202, 'Commander');
INSERT INTO Participation VALUES (1004, 203, 'Naval Officer');
SELECT * FROM Participation;

COMMIT;

SAVEPOINT before_update

UPDATE Soldier SET age = 26 WHERE soldier_id = 1001;

ROLLBACK TO before_update;    -- undo update

DELETE FROM Soldier WHERE soldier_id = 1002;
ROLLBACK;    -- undo delete
```

Results Explain Describe Saved SQL History

0.00 seconds

- **UPDATE COMMAND USED**

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```
INSERT INTO Participation VALUES (1001, 201, 'Infantry');
INSERT INTO Participation VALUES (1002, 201, 'Medic');
INSERT INTO Participation VALUES (1003, 202, 'Commander');
INSERT INTO Participation VALUES (1004, 203, 'Naval Officer');
SELECT * FROM Participation;

COMMIT;

SAVEPOINT before_update

UPDATE Soldier SET age = 26 WHERE soldier_id = 1001;

ROLLBACK TO before_update;   -- undo update

DELETE FROM Soldier WHERE soldier_id = 1002;
ROLLBACK;   -- undo delete
```

Results Explain Describe Saved SQL History

1 row(s) updated.

0.00 seconds

- AFTER UPDATING

☐ Autocommit Display 10 ▾

```
CREATE TABLE Soldier
(
soldier_id  NUMBER PRIMARY KEY,
name        VARCHAR2(50) NOT NULL,
age         NUMBER CHECK (age > 18),
rank_id     NUMBER,
unit_id     NUMBER,
CONSTRAINT fk_soldier_rank FOREIGN KEY (rank_id) REFERENCES Rank(rank_id),
CONSTRAINT fk_soldier_unit FOREIGN KEY (unit_id) REFERENCES Unit(unit_id)
);

INSERT INTO Soldier VALUES (1001, 'Rahul Singh', 25, 1, 101);
INSERT INTO Soldier VALUES (1002, 'Arjun Mehta', 28, 2, 101);
INSERT INTO Soldier VALUES (1003, 'Vikram Rao', 32, 3, 102);
INSERT INTO Soldier VALUES (1004, 'Karan Kapoor', 40, 4, 103);
SELECT * FROM Soldier;
```

Results Explain Describe Saved SQL History

SOLDIER_ID	NAME	AGE	RANK_ID	UNIT_ID
1001	Rahul Singh	26	1	101
1002	Arjun Mehta	28	2	101
1003	Vikram Rao	32	3	102
1004	Karan Kapoor	40	4	103

4 rows returned in 0.00 seconds

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```
SAVEPOINT before_update;

UPDATE Soldier SET age = 26 WHERE soldier_id = 1001;

ROLLBACK TO before_update;

DELETE FROM Soldier WHERE soldier_id = 1002;

ROLLBACK;
```

Results

Explain

Describe

Saved SQL

History

0.00 seconds

- **ROLLBACK**

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```
INSERT INTO Participation VALUES (1001, 201, 'Infantry');
INSERT INTO Participation VALUES (1002, 201, 'Medic');
INSERT INTO Participation VALUES (1003, 202, 'Commander');
INSERT INTO Participation VALUES (1004, 203, 'Naval Officer');
SELECT * FROM Participation;

UPDATE Soldier SET age = 26 WHERE soldier_id = 1001;
ROLLBACK;

GRANT SELECT, INSERT ON Soldier TO trainee_user;
REVOKE INSERT ON Soldier FROM trainee_user;
```

Results

Explain

Describe

Saved SQL

History

Statement processed.

0.00 seconds

CODE:

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```
CREATE OR REPLACE PROCEDURE add_soldier
(
  p_id NUMBER, p_name VARCHAR2, p_age NUMBER, p_rank NUMBER, p_unit NUMBER
) AS
BEGIN
  INSERT INTO Soldier VALUES (p_id, p_name, p_age, p_rank, p_unit);
  COMMIT;
  DBMS_OUTPUT.PUT_LINE('Soldier added successfully: ' || p_name);
END;
/
```

Results Explain Describe Saved SQL History

Procedure created.

0.04 seconds

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```
CREATE OR REPLACE FUNCTION get_soldier_count(p_unit NUMBER) RETURN NUMBER IS
  v_count NUMBER;
BEGIN
  SELECT COUNT(*) INTO v_count FROM Soldier WHERE unit_id = p_unit;
  RETURN v_count;
END;
/
```

Results Explain Describe Saved SQL History

Function created.

0.00 seconds

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```
CREATE OR REPLACE TRIGGER trg_check_age
BEFORE INSERT OR UPDATE ON Soldier
FOR EACH ROW
BEGIN
IF :NEW.age < 18 THEN
RAISE_APPLICATION_ERROR(-20001, 'Soldier age must be at least 18.');
```

Results Explain Describe Saved SQL History

Trigger created.

0.02 seconds

- JOINT

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```
SELECT s.soldier_id, s.name, r.rank_name, u.unit_name
FROM Soldier s
JOIN Rank r ON s.rank_id = r.rank_id
JOIN Unit u ON s.unit_id = u.unit_id;
```

```
SELECT u.unit_name, COUNT(s.soldier_id) AS soldier_count
FROM Unit u
LEFT JOIN Soldier s ON u.unit_id = s.unit_id
GROUP BY u.unit_name;
```

Results Explain Describe Saved SQL History

SOLDIER_ID	NAME	RANK_NAME	UNIT_NAME
1001	Rahul Singh	Private	Northern Command
1002	Arjun Mehta	Sergeant	Northern Command
1003	Vikram Rao	Captain	Eastern Command
1004	Karan Kapoor	Major	Naval Fleet Alpha

4 rows returned in 0.00 seconds

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```
SELECT u.unit_name, COUNT(s.soldier_id) AS soldier_count
FROM Unit u
LEFT JOIN Soldier s ON u.unit_id = s.unit_id
GROUP BY u.unit_name;
```

```
SELECT s.name, p.role
FROM Soldier s
JOIN Participation p ON s.soldier_id = p.soldier_id
JOIN Operation o ON p.operation_id = o.operation_id
WHERE o.operation_name = 'Operation Vijay';
```

Results Explain Describe Saved SQL History

UNIT_NAME	SOLDIER_COUNT
Northern Command	2
Eastern Command	1
Naval Fleet Alpha	1

3 rows returned in 0.00 seconds

[CSV Export](#)

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```
SELECT u.unit_name, COUNT(s.soldier_id) AS soldier_count
FROM Unit u
LEFT JOIN Soldier s ON u.unit_id = s.unit_id
GROUP BY u.unit_name;
```

```
SELECT s.name, p.role
FROM Soldier s
JOIN Participation p ON s.soldier_id = p.soldier_id
JOIN Operation o ON p.operation_id = o.operation_id
WHERE o.operation_name = 'Operation Vijay';
```

Results Explain Describe Saved SQL History

NAME	ROLE
Rahul Singh	Infantry
Arjun Mehta	Medic

2 rows returned in 0.00 seconds

[CSV Export](#)

- DROP TABLE

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```
DROP TABLE Participation CASCADE CONSTRAINTS;
```

```
CREATE TABLE Rank
(
rank_id      NUMBER PRIMARY KEY,
rank_name    VARCHAR2(50) NOT NULL
);
```

```
INSERT INTO Rank VALUES (1, 'Private');
INSERT INTO Rank VALUES (2, 'Sergeant');
INSERT INTO Rank VALUES (3, 'Captain');
INSERT INTO Rank VALUES (4, 'Major');
```

Results Explain Describe Saved SQL History

Table dropped.

0.13 seconds

☒ Autocommit Display 10 ▼

```
CREATE TABLE Participation
(
soldier_id    NUMBER,
operation_id   NUMBER,
role          VARCHAR2(50),
CONSTRAINT pk_participation PRIMARY KEY (soldier_id, operation_id),
CONSTRAINT fk_participation_soldier FOREIGN KEY (soldier_id) REFERENCES Soldier(soldier_id),
CONSTRAINT fk_participation_operation FOREIGN KEY (operation_id) REFERENCES Operation(operation_id)
);
```

```
INSERT INTO Participation VALUES (1001, 201, 'Infantry');
INSERT INTO Participation VALUES (1002, 201, 'Medic');
INSERT INTO Participation VALUES (1003, 202, 'Commander');
INSERT INTO Participation VALUES (1004, 203, 'Naval Officer');
```

```
SELECT * FROM Participation;
```

Results Explain Describe Saved SQL History

no data found

5. Skill Developed/Learning out of this Project

Database Design: Learned to create entity-relationship diagrams and design normalized database schemas.

Oracle SQL & PL/SQL: Gained practical experience writing complex queries, triggers, and stored procedures.

Data Security: Understood role-based access control and data encryption techniques within Oracle.

Project Management: Developed skills in planning, organizing, and implementing a real-world database project.

Problem-Solving: Addressed challenges in data integrity, concurrency, and efficient query design.

Report Generation: Automated the generation of detailed reports useful for defence operations.

Technical Writing: Improved documentation skills by preparing detailed project reports and manuals.